

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0449

Electric 100 kV
January 22, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0449. The Interconnection Customer proposes a 90.9 MW battery energy storage facility interconnecting at the Electric 100 kV substation in the Illinois area. The request was received on March 23, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 196 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$870,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Shell Chemical-Electric 100 kV Cir. 1
Contingency: Electric Substation-Shell Chemical 100 kV Circuit 2
Loading: 104.7% (Rating: 141 MVA)
DFAX: 87.5%
Pre-project loading: 48.4%
2. Facility: Electric-Shell Chemical 100 kV Ckt 2
Contingency: Electric Substation-Shell Chemical 100 kV
Loading: 104.7% (Rating: 141 MVA)
DFAX: 87.5%
Pre-project loading: 48.4%
3. Facility: Revere Dr-Corona Ae 100 kV Ckt 1
Contingency: Revere Dr Substation-Corona Ave Substation 100 kV line
Loading: 101.2% (Rating: 154 MVA)
DFAX: 10.1%
Pre-project loading: 95.3%
4. Facility: Corona Ave-Reveredr 100 kV Circuit 2
Contingency: Revere Dr-Coron Ave 100 kV
Loading: 101.2% (Rating: 154 MVA)
DFAX: 10.1%
Pre-project loading: 95.3%
5. Facility: Corona Ave-Revere Dr 100 kV
Contingency: Revere Dr-Coron Ave Substation 100 kV
Loading: 101.2% (Rating: 154 MVA)
DFAX: 10.1%
Pre-project loading: 95.3%

6. Facility: Sarival-Bpa Troutdale 100 kV Line
 Contingency: Brenas 9201-Sarival Substation 100 kV
 Loading: 100.9% (Rating: 121 MVA)
 DFAX: 8.0%
 Pre-project loading: 95.0%
7. Facility: Revere Dr-Acec Turtle Lake 100 kV
 Contingency: Bedford Park-Oro Loma 100 kV
 Loading: 100.4% (Rating: 135 MVA)
 DFAX: 10.9%
 Pre-project loading: 93.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Shell Chemical-Electric 100 kV Cir. 1	Reconductor line	\$324,000	30% (\$98,000)
Electric-Shell Chemical 100 kV Ckt 2	Reconductor line	\$415,000	30% (\$125,000)
Corona Ave-Revere Dr 100 kV Ckt 1	Reconductor line	\$833,000	23% (\$193,000)
Revere Dr-Corona Ave 100 kV	Reconductor line	\$454,000	23% (\$105,000)
Revere Dr-Corona Ave 100 kV line	Reconductor line	\$794,000	23% (\$184,000)
Bpa Troutdale-Sarival 100 kV L	Reconductor line	\$592,000	9% (\$55,000)
Revere Dr-Acec Turtle Lake 100 kV	Reconductor line	\$475,000	23% (\$110,000)
	Total allocated to project		\$870,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.